

Surrogate model assisted multi-criteria operation evaluation of community integrated energy systems

Jintao Jiang^a, Hao Yu^{a,*}, Guanyu Song^a, Jinli Zhao^a, Kunpeng Zhao^b, Haoran Ji^a, Peng Li^a

^a Key Laboratory of Smart Grid of Ministry of Education, Tianjin University, Tianjin 300072, China

^b State Grid Customer Service Center, Tianjin 300300, China

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ABSTRACT

Operational evaluation is critical for revealing the operational performance and improvement potential of community integrated energy systems (CIESs). However, complex system configuration and flexible energy conversion in a CIES render the evaluation more challenging. Both the loss in energy quality and quantity should be considered to comprehensively evaluate the energy utilization capability of a CIES. Accurately quantified evaluation requires the optimal solution as the reference, which incurs a heavy computational burden and limits its applicability under massive scenarios. This paper proposes a surrogate model assisted quantified evaluation method for CIES operation considering the variation in energy quality and to show the gap to optimal performance. First, exergy theory is adopted to analyze the matching degree between the input and output energy quality. Second, three indicators are considered as examples to reveal the operational performance in terms of system efficiency, renewable energy penetration, and operation cost. The scores of quantified evaluation are derived using the technique for order preference by similarity to an ideal solution (TOPSIS) method. Third, a CNN-based surrogate model is designed to replace the time-consuming optimization problems for the optimal reference solution, which effectively accelerates the evaluation efficiency under massive scenarios. Case studies demonstrate the effectiveness of the evaluation methods and efficiency of the proposed surrogate model.

Introduction

Increasing concerns on the global energy crisis and environmental issues have promoted interest in clean, reliable, efficient, and economic energy technologies [1,2]. The integrated energy system (IES) opens new possibilities for integrating various energy sources, such as natural gas, electricity, and heating energy [3,4]. It provides low-carbon, flexible, and efficient energy supplies via the synergistic interaction of multiple heterogeneous energy subsystems [5]. The community integrated energy system (CIES) is an important prototype of IES applications, which constructs a direct pathway to connect the end-users with the energy generation and conversion systems [6,7]. However, unlike in energy systems with single energy carriers, to evaluate the performance of a CIES is more complicated. Generally, multiple routes and devices can be chosen to satisfy the user demands in a CIES. Different system configurations and operation modes can be designed and optimized. These properties make it difficult to objectively assess the performance, benefits, and potential of a CIES.

Background and motivation

The operational evaluation of a CIES is intended to quantitatively evaluate the operational performance in terms of economy [8], sustainability [9], reliability [10,11], and efficiency [12]. The evaluation results are essential for the implementation of CIES technologies at different stages, including planning, dispatch, maintenance, and upgrading. For instance, the evaluation results of operational strategies before their implementation can be used as a reference for optimized decisions. The adaptability of system configurations in various scenarios can be analyzed by performing technical and economic evaluation of energy systems, which enriches the knowledge about system planning and design [13].

Challenges remain in the operational evaluation of the CIES. The first challenge is how to consider the differences in energy quality in addition to energy quantity in the efficiency evaluation of a CIES. Efficiency analysis is typically performed in an energy system based on the principle of energy conservation [14,15]. Only the quantity of energy loss is considered in the efficiency index, whereas the loss of energy quality is ignored. However, when high-quality energy, such as electricity, is used

* Corresponding author.

E-mail address: tjuyh@tju.edu.cn (H. Yu).

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Nomenclature			
Abbreviations			
IES	Integrated energy system	E^{EL}	Exergy of consumers' electricity loads
CIES	Community integrated energy system	E^{Gas}	Exergy of the gas fuel
CHP	Combined heat and power	F^{Gas}	Consumption of gas fuel
CCHP	Combined cooling, heating and power	$T_{\text{Esw}}, T_{\text{Erw}}$	Temperatures of the working fluid inlet and outlet of the evaporator
CNN	Convolutional neural network	E^{Ground}	Exergy consumed by geo-heating
TOPSIS	Technique for order preference by similarity to an ideal solution	Q^{GSHP}	Energy generated by the GSHP in the heating state
DG	Distributed generation	$T_{\text{h}}, T_{\text{c}}$	Terminal temperature of working fluid of heating and cooling
PV	Photovoltaics	$E^{\text{HL}}, E^{\text{CL}}$	Exergy of the heating and cooling loads
GSHP	Ground source heat pump	$Q^{\text{HL}}, Q^{\text{CL}}$	Energy of the heating and cooling loads
MAPE	Mean absolute percentage error	η_{SE}	Indicator of system efficiency
Sets		η_{REP}	Indicator of renewable energy penetration
x	Set of the source side of electricity	C_{OC}	Indicator of operation cost
k	Set of the cost types of system	$\sum R_{\text{in}}$	Total exergy of renewable energy provided by the supply side
Ω	Set of the energy forms	$\sum R_{\text{end}}$	Total exergy of renewable energy consumed by the user side
Indices		$\sum U_{\text{end}}$	Total exergy of non-renewable energy consumed the user side
i	Index of operation samples	C^k	Types of cost from set k considered in operation
j	Index of indicator types	D, D'	Decision matrix and normalized decision matrix
Parameters		r_{ij}, r'_{ij}	Indicator and normalized indicator j of operation sample i
c_w	Work fluid specific heat capacity	ω_j	Weight of indicator j
ρ_w	Density of work fluid	d_i^+, d_i^-	Distance between the sample i to the positive ideal solution and the negative ideal solution
V_w	Volumetric flow rate of work fluid	R_i	Relative closeness of sample i to the ideal solution
ψ^{Gas}	Conversion coefficient of exergy of gas fuel	S_i	Evaluation score of sample i
L^{Gas}	Lower heating value of gas fuel	$Q_{i,t}^{\Omega}$	Load of operation sample i at time t including electricity, heating and cooling
T_0	Ambient temperature	$\lambda_i^{\text{EH}}, \lambda_i^{\text{EC}}$	Daily electrical heating ratio and the daily electrical cooling ratio of sample i
T	Time period	$\xi_{i,t}^{\Omega}$	Load distribution of sample i at time t including electricity, heating and cooling
Variables		y_i, γ_i	Indicator values solved by surrogate model and optimization of sample i
$E_{\text{in}}^{\text{Grid}}$	Exergy generated from external power grid	M_{APE}	Mean absolute percentage error
$E_{\text{in}}^{\text{DG}}$	Exergy generated from distributed generation	M_{A}	Mean accuracy
$E_{\text{in}}^{\text{GF}}$	Exergy of gas fuels from the external gas grids	a_j	Weight of optimal indicator j
$E_{\text{in}}^{\text{AH}}$	Exergy generated from ambient environment	$M_{\text{A},j}$	Mean accuracy of optimal indicator j
$E_{\text{out}}^{\text{PL}}$	Exergy of the terminal electrical loads	W_{MA}	Weighted mean accuracy of the surrogate model
$E_{\text{out}}^{\text{HL}}$	Exergy of the terminal heating loads		
$E_{\text{out}}^{\text{CL}}$	Exergy of the terminal cooling loads		
E_{loss}	Exergy lost in the process of energy transfer and conversion		
P, E	Energy and exergy of an energy carrier		
p^{EL}	Consumers' electricity demand		

to meet low-quality demands, such as indoor heating, the deterioration of energy quality should also be regarded as a waste of resources that decreases the efficiency performance of the system [16].

The second challenge is to quantitatively reflect the gap between the operation sample and optimal performance of the evaluation results. The distance to the optimal performance is particularly important in the pre-evaluation of operation strategies and post-evaluation of historical operation data [17]. In the pre-evaluation with numerous candidate operation strategies, quantified scores can indicate proximity to the optimal performance and, thus, accelerate the decision process during pre-scheduling. For the post-evaluation of large-scale historical data, the quantified scores that indicate the distance to the optimal performance can be used to discover typical days and unusual scenarios, which is helpful for exploring the system's improvement potential. However, the distance to the optimal performance can only be determined when the optimal solution is known, and its calculation relies on solving very time-consuming optimization problems. This situation is worse for multi-objective problems. Moreover, both pre-evaluation and post-

evaluation problems require the calculation of numerous scenarios, which further increases the computational burden.

Literature review

Extensive studies on the operational evaluation methods of IES and CIES have been conducted. It can be concluded from existing studies that the economy, sustainability and efficiency are the most concerning objectives in the evaluation of energy systems. For example, ref. [8] proposed a multi-objective optimization and multi-criteria evaluation framework to guide the planning and operation of distributed energy systems. Ref. [12] developed an evaluation framework that considered five aspects, including economy, environment, reliability, sustainability, and energy utilization, providing comprehensive and quantified information support for CIES decision-making. Ref. [18] analyzed the characteristics of combined heat and power systems in different buildings in China and Japan and ranked the system performance using a multi-criteria evaluation method. Ref. [19] evaluated the performance of

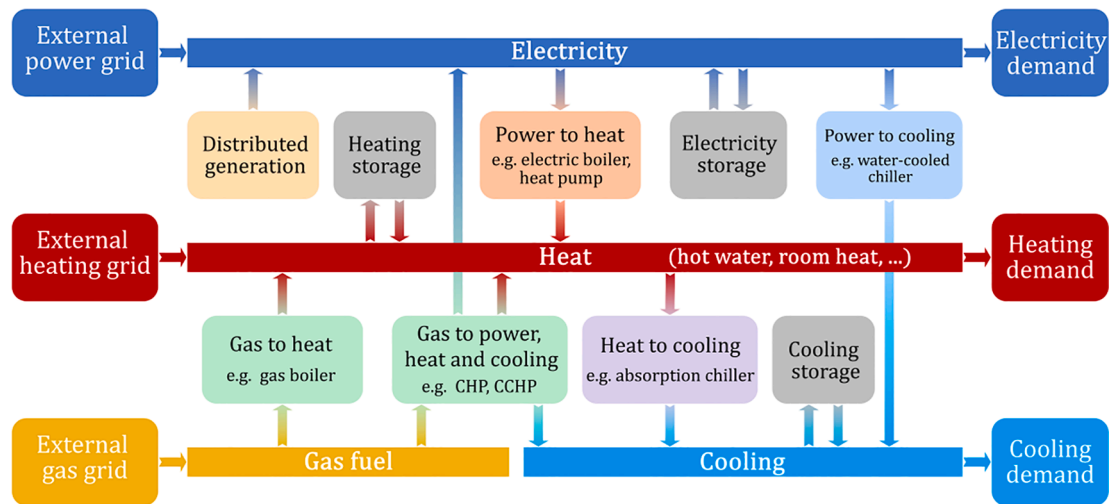


Fig. 1. Schematic for the energy relationships in a CIES.

three different combined heating and power (CHP) plant layout options to determine the best choice for the short term. Ref. [20] proposed a multi-criteria method to choose the optimal renewable energy sources (RESs) in industrial sector considering expert opinions, the costs of operation and maintenance, and the electricity from RES. An evaluation method for the foremost window allocation scheme was proposed in [21] to obtain the best trade-off among energy, environmental, and comfort criteria in a building.

Exergy-based methods have been developed to address the loss of energy quality in the efficiency analysis of a complex system. Exergy is defined as the maximum work that can be produced by a stream or system in a specified environment, which can be utilized to quantitatively measure of the quality or usefulness of a given amount of energy [22,23]. For example, ref. [24] presented an exergy analysis of a 650 MW thermal power plant. Ref. [25] applied exergy analysis to evaluate the operational flexibility of the primary frequency process in coal-fired power plants. Ref. [26] evaluated a renewable energy source by considering exergy efficiency, including solar and wind power, hydrogen production units, and fuel cells. Energy and exergy analysis methods were applied in [27] to explore the tunable performance of a combined cooling, heating, and power (CCHP) system with variable solar irradiance and solar thermal allocation ratios. The principle of exergy can be utilized as a possible tool for evaluating the efficiency of a CIES with flexible energy conversion.

The surrogate model methods are considered effective methods of reducing the computational burden in complex systems. The approach of the surrogate model is derived from the multidisciplinary design sector, and it can effectively deal with complex and time-consuming optimization problems and then to various optimization sectors [28]. Surrogate models generally utilize data-driven approaches to abstract the original problem into a black-box model. For example, in [29], a modelling strategy using the Kriging surrogate model for aerodynamics was proposed. Ref. [30] used a surrogate model of a support vector machine to explore the influence of weather conditions on the design optimization of high-rise residential buildings. Ref. [31] used a convolutional neural network (CNN) to construct a surrogate model for power systems to predict the time-history response of transmission towers during complex wind scenarios. The application of surrogate models not only improves the efficiency of optimization, but also reduces solving difficulty. This provides a promising method to accelerate complicated optimization in the quantified evaluation of CIES.

Our contribution

This paper proposes a novel quantified operation evaluation method

for CIESs considering the variation in energy quality and the gap between the system and optimal performance. Three indicators are considered as examples to reflect the key operational benefits of the CIES: system efficiency, renewable energy penetration, and operation cost. The evaluation scores are derived from the technique for order preference by similarity to an ideal solution (TOPSIS) method, whose calculation is further accelerated by using a CNN-based surrogate model for enhanced evaluation efficiency under massive scenarios. The main contributions of this study are summarized as follows:

1) Exergy models are constructed for the efficiency evaluation of the CIES, considering the variations in energy quality. The electrical, gas, cooling, and heating energy, as well as energy conversion devices, are incorporated in a black-box to indicate the exergy balance relationship. The reasonability of the evaluation results regarding the efficiency performance is improved.

2) A comprehensive operation evaluation method is developed for CIES considering efficiency, sustainability, and economic performance. The TOPSIS method is used to discover the gap between the samples and the optimal performance and formulate the quantified scores, which provide an effective reference for operational analysis and decision-making in a CIES.

3) A CNN-based surrogate model is developed to accelerate the calculation of the quantified evaluation. The surrogate model is pre-trained and capable of directly generating optimal indicators for any given scenario, which avoids intensive optimization during evaluation. The surrogate model enabled the application of the proposed evaluation method to massive scenarios.

The remainder of this paper is organized as follows. Section 2 introduces the detailed exergy-based modelling method of the CIES. Section 3 presents an operational evaluation process based on the TOPSIS method. Section 4 proposes a surrogate model designed to accelerate the calculation of indicators for massive scenarios. Case studies to verify the effectiveness of the proposed method are presented in Section 5. The conclusions are presented in Section 6.

Modelling for CIES operation evaluation

In this section, the universal exergy-based modelling method is presented for the operation evaluation of CIES. Three indicators are selected to evaluate the performance in aspects of system efficiency, sustainability and economy.

Exergy-based analysis of CIES

A CIES generally covers the generation, conversion, and utilization of

different forms of energy, such as electricity, natural gas, and heat and cooling energy, as shown in Fig. 1. The exergy analysis method was developed based on the second law of thermodynamics [27]. It extends the conventional energy analysis method and describes the energy conversion and utilization processes in terms of both quantity and quality. The exergy has the same unit of measurement as the energy but has different grades for various energy forms to indicate the differences in energy quality. Hence, the exergy analysis can reflect the matching degree of the energy quality levels between supply and demand, which is also an important indicator for evaluating the efficiency performance of a CIES.

The energy consumed in a CIES is provided by multiple sources, such as the external grid, external gas grid, distributed generation (DG), and air-heat or geo-heat energy. For the exergy-based efficiency analysis of a CIES, the entire system is regarded as a black box, as shown in Fig. A.1 in Appendix A. According to the second law of thermodynamics, the input and output exergy flows of the system satisfy the following balanced equation:

$$E_{in}^{Grid} + E_{in}^{DG} + E_{in}^{GF} + E_{in}^{AH} = E_{out}^{PL} + E_{out}^{HL} + E_{out}^{CL} + E_{loss} \quad (1)$$

where E_{in}^{Grid} is the power exergy from the external power grid; E_{in}^{DG} is the input exergy generated by distributed generators, such as photovoltaics (PVs) and wind turbines; E_{in}^{GF} is the exergy of gas fuels from the external gas grids; E_{in}^{AH} is the exergy from the ambient environment, such as air or geo; E_{out}^{PL} , E_{out}^{HL} , and E_{out}^{CL} are the exergies of the terminal electrical loads, heating loads, and cooling loads, respectively; E_{loss} is the exergy lost in the process of energy transfer and conversion within the system. According to engineering thermodynamics theory, the exergy variables in (1) can all be determined based on the quantity and quality factor of the corresponding energy form [27,32].

Modelling of the exergy relationships in a CIES

1) Exergy-based modelling of electricity.

Electricity has the highest exergy grade because it can be converted entirely into other forms of energy. Hence, the exergy of electricity is assumed to be numerically equal to the amount of energy. For a CIES, despite the electricity from the external grid, the energy from some renewables is also utilized as electricity via distributed generation techniques, such as PVs and wind turbines. Therefore, the exergy from solar and wind energy is calculated in the same manner as the electricity from the external power grid. The exergy calculation of electricity is written as:

$$E^x = P^x, \quad x \in \{\text{Grid}, \text{DG}\} \quad (2)$$

where P represents the energy of electricity; E represents the exergy of electricity; $x \in \{\text{Grid}, \text{DG}\}$ represents the source of electricity.

With the same assumption as that for the electricity supply, the exergy of the electricity consumed by users in a CIES can be calculated as follows:

$$E^{EL} = P^{EL} \quad (3)$$

where P^{EL} is the consumers' electricity demand; E^{EL} is the exergy of the electricity loads.

2) Exergy-based modelling of gas fuel.

The exergy of gas fuel (natural gas, biomass, hydrogen, etc.) belongs to the category of chemical exergy, defined as the maximum work obtained from a substance when it reaches chemical equilibrium with a reference temperature at constant temperature and pressure [16]. The exergy of the gas fuel is calculated as follows:

$$E^{Gas} = \psi^{Gas} L^{Gas} F^{Gas} \quad (4)$$

where ψ^{Gas} is the conversion coefficient of exergy [22], which is usually

considered as a constant [16,33]; L^{Gas} is the lower heating value of the gas fuel; F^{Gas} is the consumption of gas fuel; E^{Gas} is the exergy of the gas fuel.

3) Exergy-based modelling of heat and cooling energy.

The supply of thermal energy can be classified into two categories in a CIES. The first is the generation of heat energy via electrical or gas combustion. The exergy of these processes is considered in the exergy of the electricity or gas consumed. The second is heat pumps. The exergy extracted from the external environment by the heat pumps can be calculated using the temperature of the working fluid flowing into and out of the equipment. Taking the ground source heat pump (GSHP) as an example, in winter, it extracts thermal energy from the earth and transfers the energy to the user side through the circulation of the working fluid inside the heat pump. Therefore, the geo-heating exergy can be regarded as the energy absorbed by the heat exchange of the fluid in the evaporator on the ground source side, which can be calculated as follows [33]:

$$E^{Ground} = c_w \rho_w V_w \left[(T_{Esw} - T_{Erw}) - T_0 \ln \frac{T_{Esw}}{T_{Erw}} \right] \quad (5)$$

where c_w , ρ_w , and V_w are the specific heat capacity, density, and volumetric flow rate of the working fluid, respectively; E^{Ground} is the exergy consumed during geo-heating; T_0 is the ambient temperature; T_{Esw} and T_{Erw} are the temperatures of the working fluid inlet and outlet of the evaporator, respectively.

Under constant temperature and pressure conditions, the working fluid flow rates at the condenser and evaporator sides of the GSHP are approximately the same. Thus, the geo-heating exergy can be simplified as follows:

$$E^{Ground} \approx Q^{GSHP} \left(1 - \frac{T_0}{T_{Esw} - T_{Erw}} \ln \frac{T_{Esw}}{T_{Erw}} \right) \quad (6)$$

where Q^{GSHP} denotes the energy generated by the GSHP in the heating state. The same process applies to the calculation of exergy in the cooling mode and can be extended to similar heat pump devices, such as air-source heat pumps [33].

The exergy consumed by the heating or cooling loads depends on the ratio of the working fluid temperature T_h/T_c and ambient temperature T_0 , which are expressed as follows [34]:

$$E^{HL} = Q^{HL} \left(1 - \frac{T_0}{T_h} \right) \quad (7)$$

$$E^{CL} = Q^{CL} \left(\frac{T_0}{T_c} - 1 \right) \quad (8)$$

where Q^{HL} and Q^{CL} are the heating and cooling energy demands at the user side, respectively; E^{HL} and E^{CL} are the exergies of the heating and cooling loads, respectively.

Indices definition and calculation

To comprehensively evaluate the operational performance of the CIES, three typical indices are considered in this study: system efficiency, renewable energy penetration, and operation cost. Definitions for the three indicators are explained in this section, and the detailed calculation methods are elaborated in Appendix B Equations (B.1)-(B.3). Other indices can be incorporated into the proposed evaluation method in a similar way.

1) System efficiency.

System efficiency is a measure of the energy utilization performance of a CIES by considering both energy quantity and quality. System efficiency refers to the ratio of the total exergy output to the total exergy input, which comprehensively expresses the energy utilization efficiency and matching degree of energy grades in the system.

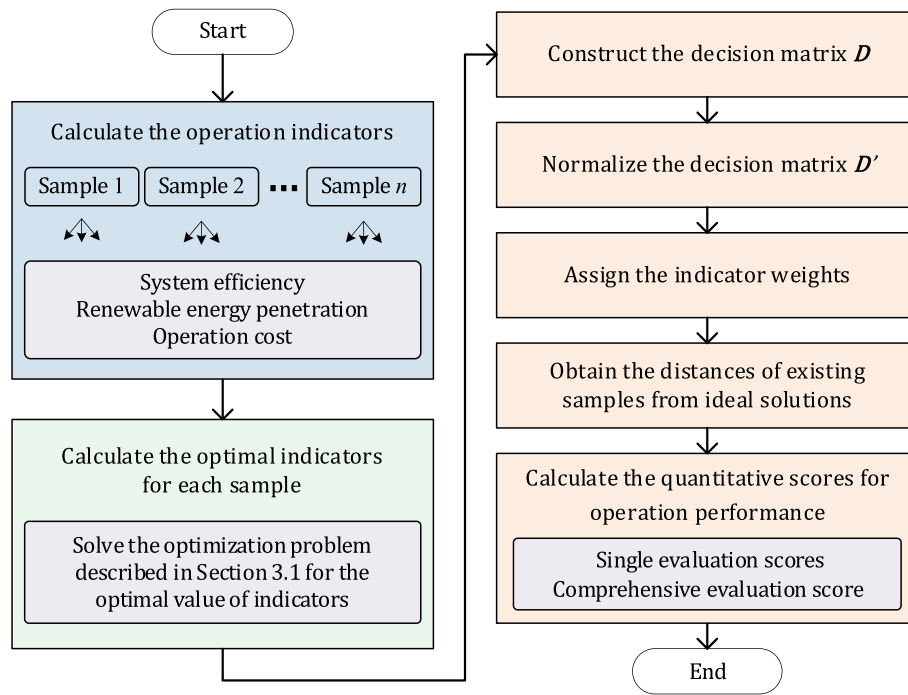


Fig. 2. Quantified evaluation process based on modified TOPSIS method.

2) Renewable energy penetration.

The environmental benefits of a CIES can be evaluated based on its ability to utilize renewable energy resources. A larger proportion of renewables in a CIES means less fossil energy consumption. Therefore, renewable energy penetration is selected as the evaluation indicator to quantify the environmentally friendly benefits. Renewable energy penetration refers to the ratio of renewable energy available on the energy supply side to the energy demand on the user side.

3) Operation cost.

The operation cost of a CIES is the most intuitive index for evaluating economic performance. In this respect, the operation cost is selected as the evaluation indicator to quantify the economics of the system operation. The operational cost refers to the economic cost incurred during the operation of the CIES.

Quantified scoring of CIES operation performance

In this section, the TOPSIS method is employed to obtain the comprehensive evaluation scores. To improve the scoring accuracy, the optimal values of indicators are utilized as references in the modified TOPSIS process.

Determination of the optimal indicators

It is critical to obtain optimal indicators as a reference for judging the evaluation samples, such as candidate operation strategies or historical operation data. It is also necessary to quantify the gap between the samples and determine the optimal performance to make the evaluation results more intuitive. The optimal indicators can be obtained by solving the optimization problems with objective functions formulated using the desired indices introduced in Section 2.3. The operation optimization of a CIES is a constrained problem, in which various constraints need to be considered, such as energy balance, equipment capacity, energy storage, and energy coupling and conversion constraints [6]. The optimal indicators for the current operation sample can then be obtained by sequentially solving the optimization problems.

Quantified scoring using TOPSIS method

TOPSIS is based on the ideal solution distance and is widely used in multi-criteria decision-making problems [35]. The classical TOPSIS method considers that the best alternative should have the shortest distance from the positive ideal solution and farthest distance from the negative ideal solution [36]. In the classic TOPSIS method, the positive and negative ideal solutions are determined as the best and worst values, respectively, among the available samples. Under this condition, the positive and negative ideal solutions are limited by the coverage of the samples, which may affect the accuracy of the evaluation ranking.

Therefore, deterministic positive and negative ideal solutions are adopted in this study to avoid the problem of insufficient samples. The optimization process in Section 3.1 can be employed to determine the positive ideal solution of the indicators, whereas the negative ideal solution is set to 0 instead of using the relatively worst solutions in the samples. This modified TOPSIS method is then used to obtain the comprehensive score of the CIES considering the economic, environmental, and exergy efficiency performance. Fig. 2 shows the evaluation and scoring processes using the modified TOPSIS method. The process is further detailed in Appendix C.

Surrogate model based on convolutional neural networks

In this section, a surrogate model is developed to replace the costly optimization process for the optimal values of indicators. The structure of the CNN-based surrogate model and the configurations of its input and output are presented.

Structure of the CNN-based surrogate model

The quantified evaluation process introduced in Section 3 involves repeatedly solving operation optimization problems, which may be large-scale mixed-integer nonlinear programming problems owing to massive scenarios, multiple objective functions, and diverse constraints [37]. Therefore, a surrogate model is designed in this study to replace the calculation of the optimal indicators for each sample shown in Fig. 2. With the surrogate model, the optimal indicators of an evaluation

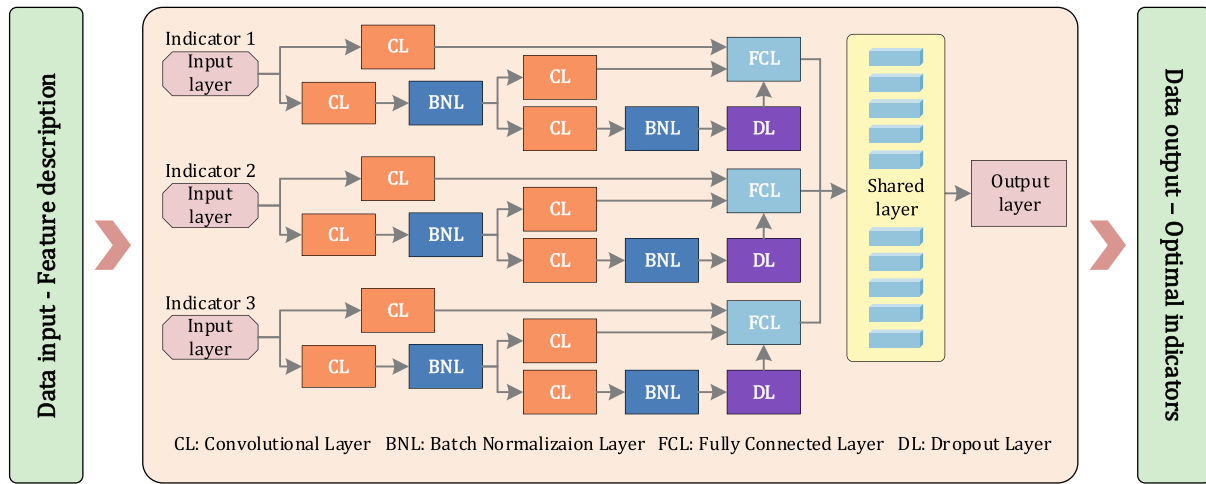


Fig. 3. Architecture of the surrogate model.

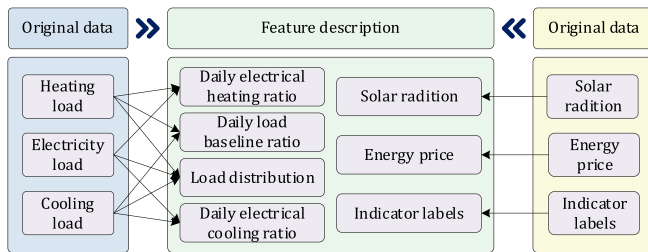


Fig. 4. Selected features that describe the operation scenario.

Table 1

Candidate scheduling strategies for the CIES.

Operation objective	Weights in each operation strategy								
	A1	A2	A3	A4	A5	A6	A7	A8	A9
min Purchase of external power	1	–	–	0.5	–	–	0.2	–	–
max Renewable energy utilization	–	1	–	–	0.5	–	–	0.2	–
min Carbon emission	–	–	1	–	–	0.5	–	–	0.2
min Operation cost	–	–	–	0.5	0.5	0.5	0.8	0.8	0.8

sample can be efficiently derived, which is very suitable for large-scale scenario applications. Considering the characteristics of different attribute data, a CNN network with a multisource input is used to construct the surrogate model. Fig. 3 shows the architecture of the surrogate model, which mainly includes three parts: the data input, internal structure of the CNN-based surrogate model, and data output.

1) Description of the data input.

The data inputted to the surrogate is a description of the operation scenario. Multiple attributes are included in the input data to fully

capture the characteristics of an operational scenario, as shown in Fig. 4. Two principles are considered in the design of the data structure. First, the input data should be capable of representing all the internal and external operating conditions, such as the daily electric/heating/cooling load profiles, energy price curve, and renewable resources. Second, the operation scenario should be expressed in a generalized pattern, and the original data should be transformed into normalized feature descriptions.

In this study, the original load profiles are transformed into the daily

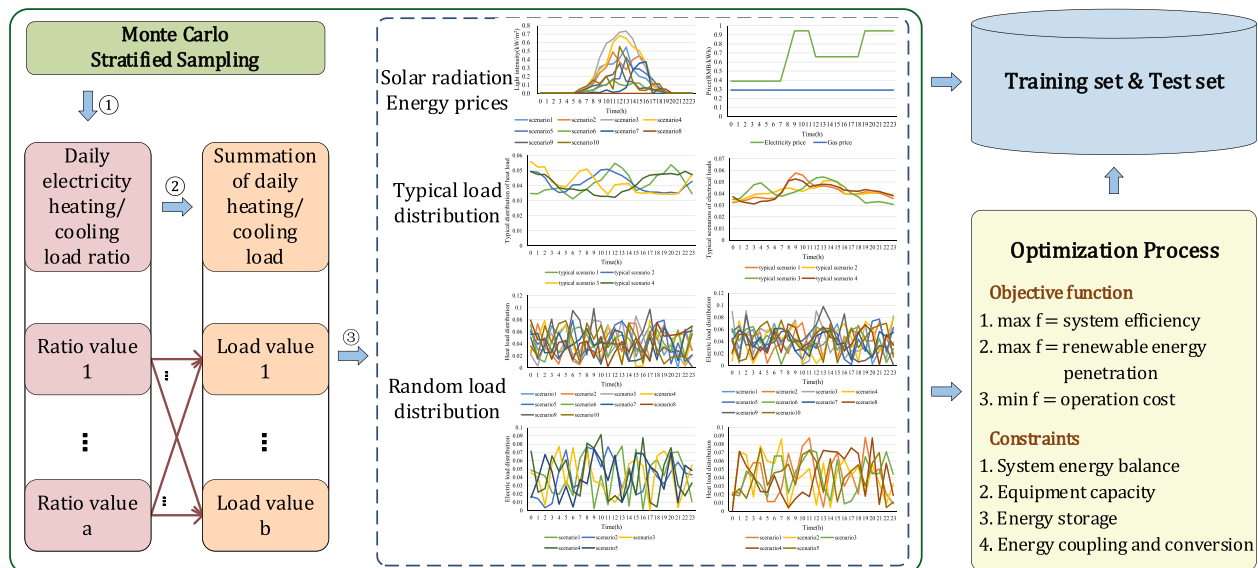


Fig. 5. Generation process of the data sets.

Table 2

Comparison of the evaluation scores of scheduling strategies A1–A9.

Strategies	Comprehensive score		Score of system efficiency		Score of renewable energy penetration		Score of operation cost	
	M1	M2	M1	M2	M1	M2	M1	M2
A1	44.75	67.41	0.00	60.04	41.58	57.74	84.95	89.47
A2	85.87	89.96	87.02	90.78	100.00	100.00	78.25	84.77
A3	37.14	49.34	84.08	89.74	0.00	27.60	0.00	30.01
A4	44.75	67.41	0.00	60.04	41.58	57.74	84.95	89.47
A5	68.18	83.68	40.74	74.43	79.70	85.37	100.00	100.00
A6	90.05	92.39	100.00	95.37	100.00	100.00	81.94	87.36
A7	55.37	75.31	16.59	65.91	59.12	70.45	97.57	98.30
A8	68.18	83.68	40.74	74.43	79.70	85.37	100.00	100.00
A9	72.94	85.58	53.08	78.79	79.70	85.37	99.16	99.41

Table 3

Performance comparison between the optimization model and surrogate model.

Performance Index	Surrogate model	Optimization model
W_{MA}	94.34 %	100.00 %
Time consumption	2.30 s	4942.94 s

electrical heating/cooling ratio, daily load baseline ratio, and load distribution to formulate a normalized feature description. The detailed calculation process is described in Appendix D. These data are pre-processed and then fed into the surrogate model through multiple sources to formulate a supervised learning problem.

2) Structure of the CNN-based surrogate model.

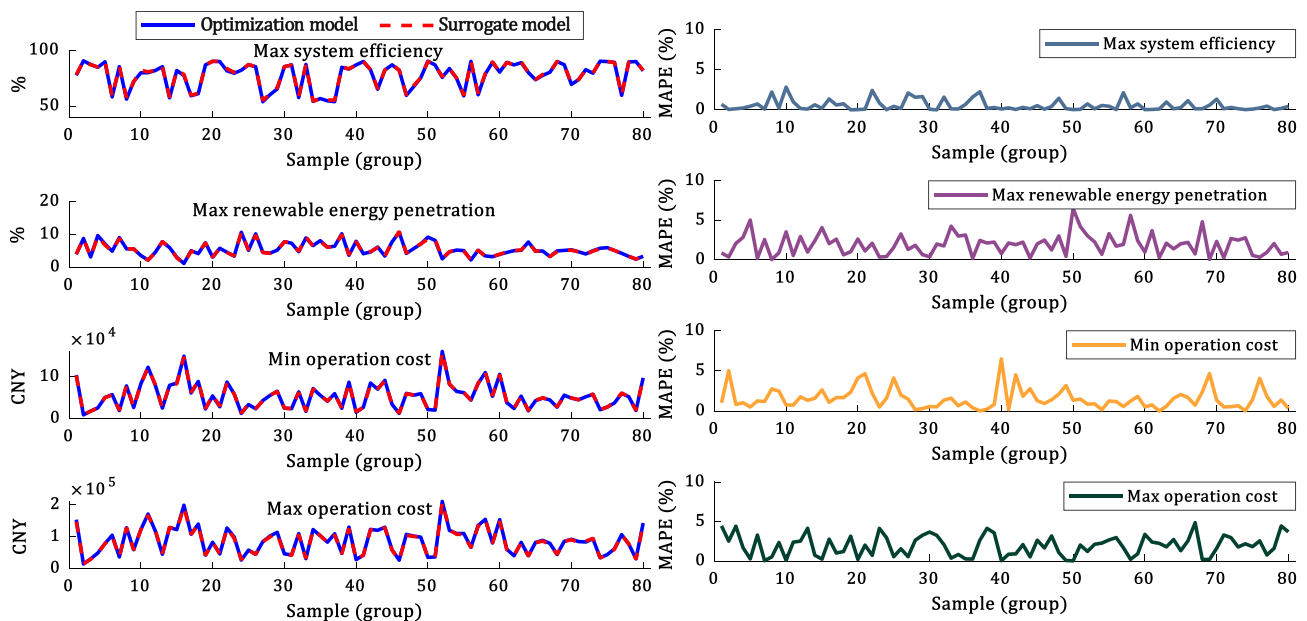
A CNN is utilized to formulate the surrogate model, which mainly consisted of five layers: the input layer, convolutional layer, batch normalization layer, dropout layer, and fully connected layer. Three branches are used to feed the sample data into the network, and each branch could be interpreted as learning for a single indicator.

The data feature is first extracted through the convolutional layer. The batch normalization layer is added to make the feature extraction more stable and accurate, which changes the output distribution of the

convolutional layer to avoid the gradient disappearance problem during training [38]. A random deactivation strategy called the dropout layer is adopted to regularize the network and reduce the probability of over-fitting [39]. The design of the residual structure of each branch enhances the network's ability to extract features and reduce network degradation [40]. The features extracted from the three branches are combined to form a new feature vector and sent to a shared layer containing multiple fully connected layers. Finally, the feature vector of optimal indicators is output.

3) Description of the data output.

The output of the surrogate model contains the optimal indicators under the input scenario. In this study, there are four output values for each operation sample input: the optimal system efficiency, optimal renewable energy penetration, optimal operation cost, and maximal operation cost. The maximum operation cost is the worst economic performance in massive scenarios and is utilized as a reference to indicate the economic improvement. For each sample, the first three output values represent the positive solutions for the three indicators, whereas the fourth value represents the negative solution for operational cost. Hence, with the proposed surrogate model, all optimal indicators of an operational sample can be directly obtained without solving time-consuming optimization problems.

**Fig. 6.** Results comparison using the optimization model and the surrogate model.

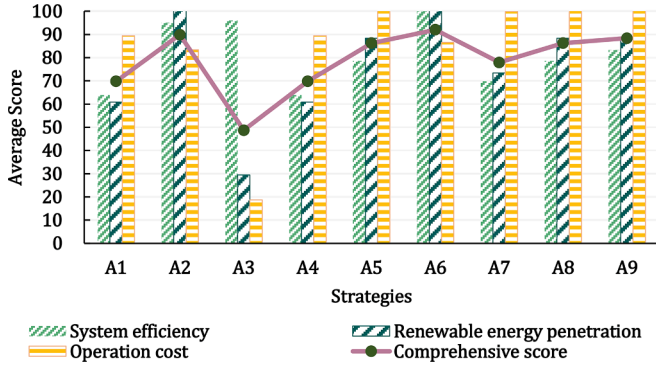


Fig. 7. Scoring results for scheduling strategies.

Training of the surrogate model

1) Generation of the data sets.

A four-stage Monte Carlo method is used to perform stratified sampling on the operational scenarios. The first stage sampled the daily electrical heating/cooling load ratio, and the second stage sampled the aggregated amount of daily heating/cooling loads. In the third stage, the sampled aggregated data are distributed into daily curves according to the two distribution properties. The first type of distribution is a typical distribution obtained from historical operation data, which is designed to enhance adaptation to common scenarios. The second type of distribution is a random distribution, which is utilized to improve the generalization ability. In the fourth stage, labels are then generated for the sampled scenarios via the optimization process, in which the three indicators of system efficiency, renewable energy penetration, and operation cost are taken as the objective functions. The entire process of generating input data for the training and test sets is illustrated in Fig. 5.

2) Training of the surrogate model.

It is necessary to set hyperparameters in the supervised learning network training. The model parameters are designed using the method introduced in [39]. The loss function of the model adopts the mean-square error function. Moreover, to measure the learning accuracy for the optimal indicators, the weighted mean accuracy W_{MA} proposed in [41] is used as the evaluation criteria, which is derived from the mean absolute percentage error M_{APE} and mean accuracy M_A . The detailed calculation process is shown in Appendix E.

Case study

In this section, a practical CIES in North China [6] is selected for this case study. Fig. A.2 in Appendix A shows the configuration of the case. The parameters of the equipment and energy prices are detailed in Table A.1 and A.2. The case study is conducted to verify the effectiveness of the operational evaluation of the CIES and the validity of the surrogate model. MatlabR2020b is used as the programming platform to build the surrogate model in the deep learning toolbox. The optimization problems involved in the case study are solved using the CPLEX solver.

Verification of the proposed evaluation method

This section conducts the pre-evaluation of various candidate operation strategies to verify the effectiveness of the proposed method. The

CIES provides different energy supply routes to meet the load demand and formulate different choices for pre-scheduling, as shown in Table 1.

As indicated in Table 1, A1–A3 are single-objective scheduling strategies considering three operational principles. A4–A6 and A7–A9 are multi-objective scheduling strategies that balance the economic and environmental objectives with different weight assignments. Employing the typical day shown in Fig. A.3 in Appendix A, the evaluation scores of different strategies are calculated and listed in Table 2. In practice, the decision-maker's preference is reflected in the weight coefficient, which affects the comprehensive score and the final decision of the strategy. The evaluation results obtained with two methods are compared in Table 2, including the classical TOPSIS method denoted by M1 and the proposed modified TOPSIS method denoted by M2.

As indicated in Table 2, when an indicator is scheduled as the operation objective, the corresponding score is the full score. For example, the score of renewable energy penetration for A2 is the highest among the three strategies, with the objective of maximizing the utilization of renewable energy. The users' preferences can be considered in the weight coefficients of optimization objectives, and thus the corresponding scores can be promoted. Comparing A2 and A5, strategy A2 considers the single objective, and strategy A5 additionally considers the economic objective, which is reflected in the score of the operation cost being higher than that in A2. It is also indicated in Table 2 that the scores of the operational strategies accurately reveal the trade-offs for different objectives. Compared to A1–A3, the score of the operation cost for A4–A6 is generally higher than that for A1–A3 because A4–A6 considers the balance between environmental and economic objectives. Compared to A4–A6, the score of operation cost is higher in A7–A9 because a more significant weight is assigned to the economic objective. The same is found for renewable energy penetration.

In the classical TOPSIS method, the positive and negative solutions are determined as the best and worst values among the sampled operational strategies. Therefore, in Table 2, there are 0 and 100 in the scores obtained with M1. Although the scores differ, the ranking of the operational strategies derived from M1 and M2 is consistent. This result indicates that the method proposed in this paper is valid and effective. Moreover, the scores derived by the proposed method reflect not only the relative relationship between the operation strategies but also the absolute distance to the optimal solution. Therefore, the result derived with M2 is more accurate and provides more valuable information, which is an advantage of the proposed method in practice.

Improvement with the surrogate method in massive scenarios

This section aims to verify the effectiveness of the surrogate model, which can be utilized to evaluate massive-scenario conditions. The surrogate model is utilized to accelerate the evaluation process of pre-scheduling with uncertainty. The method from [42] is utilized to generate random scenarios obeying a normal distribution to simulate uncertainties in practice. This is performed explicitly by introducing random errors in the typical scenario shown in Figure A.3, and 1000 sets of scenarios are sampled using Monte Carlo methods. The optimal indicators of these scenarios are then derived using the surrogate model.

1) Validity of the CNN-based surrogate model.

The training set of the surrogate model consisted of 696,000 samples obtained using the method described in Section 4.2. A total of 100,000 samples are generated for the test set. A comparison between the surrogate model and analytical optimization results is given in Table 3, showing the learning accuracy in the test set and the computational time

to solve 1000 sampled operation scenarios.

As indicated in Table 3, the weighted mean accuracy W_{MA} of the surrogate model reached 94.34 %, indicating a favorable performance in terms of accuracy. Meanwhile, it is significantly better than the optimization model in terms of time consumption, indicating that the surrogate model can effectively accelerate the evaluation process for the operation part. To further visualize the learning performance of the surrogate model, 80 samples are randomly selected from the test set, and the results are shown in Fig. 6.

As indicated in Fig. 6, the left side shows the sample indicator values solved by the optimization and surrogate models, whereas the right side shows the MAPE corresponding to each indicator of the sample. MAPE refers to the relative error between each sample value resolved using the two methods. The left side shows that the overall trends of the values calculated by the two methods are consistent. The right side reveals that the relative errors of system efficiency are all less than 5 %; therefore, system efficiency has a better learning effect than other indicators. Overall, the maximum relative errors of all four indicators are maintained within 10 %; thus, the surrogate model achieved a favorable learning accuracy.

2) Application in the pre-scheduling evaluation of massive strategies.

For the 1000 sampled scenarios accounting for uncertainty, we calculated the single and comprehensive evaluation scores in each scenario for the strategies presented in Table 1. These scores are calculated based on surrogate models to efficiently determine the optimal indicators. The average scores of each strategy are then calculated. The average scores for the massive scenarios are shown in Fig. 7.

Fig. 7 presents the average score of strategies in massive scenarios. It also indicates the average adaptability of the operation strategy to uncertain operation scenarios. This analysis shows that the surrogate model can be effectively applied for evaluating massive scenario samples. The trends in the comprehensive scores are similar to that obtained from the original operation scenario shown in Fig. A.3. Moreover, the scores can be utilized to evaluate the performance in operation scenarios with different features, which helps determine outstanding and abnormal scenarios to enhance the system knowledge and guide system improvement.

Based on the comprehensive scores in Fig. 7, all the candidate strategies are ranked from highest to lowest as A6, A2, A9, A5(A8), A7, A1 (A4), and A3. Therefore, strategy A6 is recommended to the decision-makers. Additionally, some interesting results can be observed. For example, strategies A5 and A8 have the same score in operation cost, although different weights are assigned. It indicates that the weight adjustment may be insufficient to affect the operation performance. Comparing A2 and A5, the operating cost weight is reduced to 0, and the score for operation cost reaches the lowest level. This fact indicates that the sensitivity of objective weights is a critical issue in the operation optimization of the CIES. It is also a potential application for the proposed method to determine the sensitivities of objective weights.

Conclusion

This study is an exploratory attempt focusing on CIES operation evaluation, considering the energy quality variation and the distance to the optimal performance of the system. Multidimensional operational benefits are reflected, including efficiency, sustainability, and economy.

The quantitative evaluation scores are calculated using the modified TOPSIS method. A surrogate model based on a CNN is designed to accelerate the evaluation efficiency under massive scenarios. A practical CIES in North China verifies the validity of the evaluation method and surrogate model, and the following conclusions can be drawn.

1) The evaluation method proposed in this paper can effectively reflect the performance of different operation strategies. We set nine operation strategies and compared the evaluation results using different methods, indicating that our method has advantages in accuracy because it takes the optimal indicator values as reference. Decision-makers can objectively analyze the choice of operation principle and the trade-off between multiple objectives based on the evaluation scores.

2) The CNN-based surrogate model designed in this study achieved an excellent learning accuracy and significantly reduced the computation time of the optimal indicators. The case study using the surrogate model indicates that the weighted average accuracy reaches 94.34 % and the calculation time is reduced from hours to seconds. The application of the surrogate model accelerates the evaluation process in massive scenarios.

The evaluation method proposed in this study provides a refined pre-evaluation and post-evaluation for CIES operation. The method can guide the decision of scheduling strategy based on evaluation scores and promote further CIES development. It is also useful in assessing various CIES configurations at the planning and design stage. In this study, we adopt three typical indicators as examples. Other indicators, which may be more refined or specified, can also be integrated with the proposed method to enhance its applicability in practice. Besides, the proposed surrogate model is restricted within a given CIES configuration. Additional training process is still required to evaluate different system configurations, which is computationally costly. The improvement in adaptivity of the CNN-based surrogate model can be a future research direction.

CRedit authorship contribution statement

Jintao Jiang: Data curation, Writing – original draft. **Hao Yu:** Supervision, Writing – review & editing. **Guanyu Song:** Validation, Methodology. **Jinli Zhao:** Supervision, Software. **Kunpeng Zhao:** Investigation, Project administration. **Haoran Ji:** Software, Formal analysis. **Peng Li:** Conceptualization, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

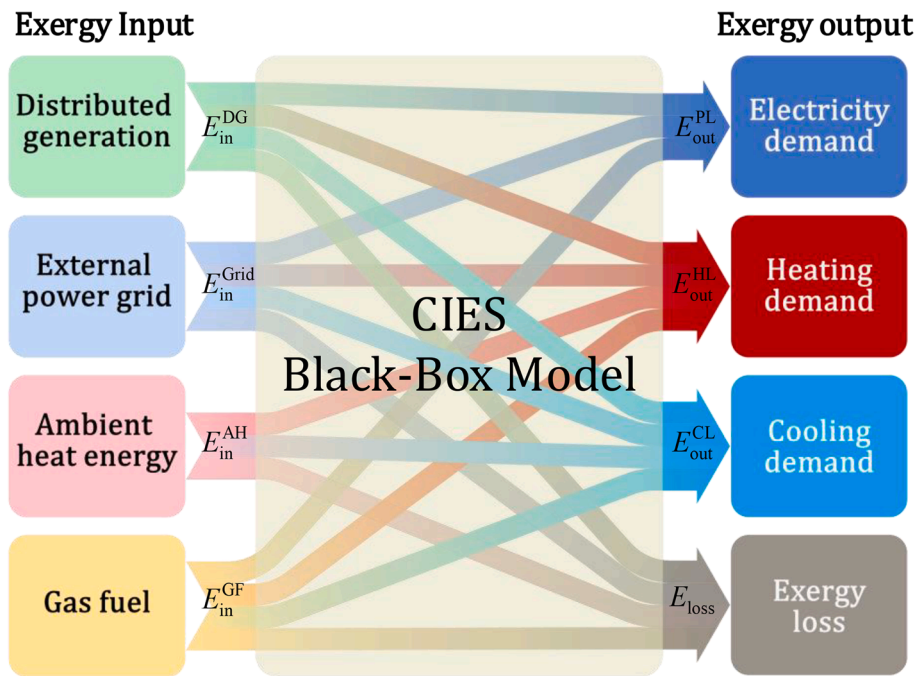


Fig. A1. Input-output black-box modelling of a CIES based on exergy analysis.

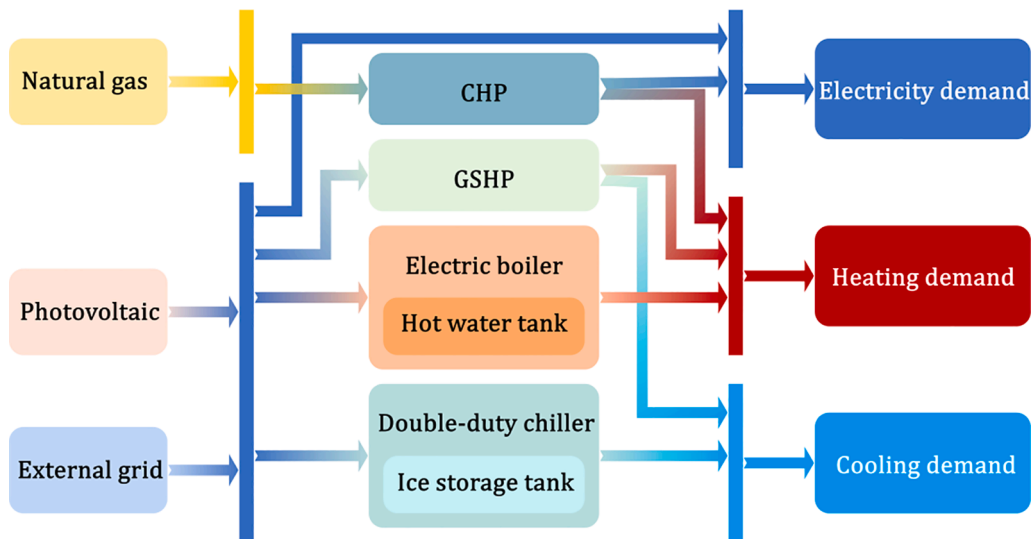


Fig. A2. Energy supply structure of the studied CIES.

The test case covers six energy forms: electricity, heating, cooling, solar energy, ambient heating energy, and natural gas. A variety of energy supply and energy storage devices, including external grid, PV, CHP, GSHP, electric boiler, and double-duty chiller, are utilized to meet the electricity, heating, and cooling loads of the CIES.

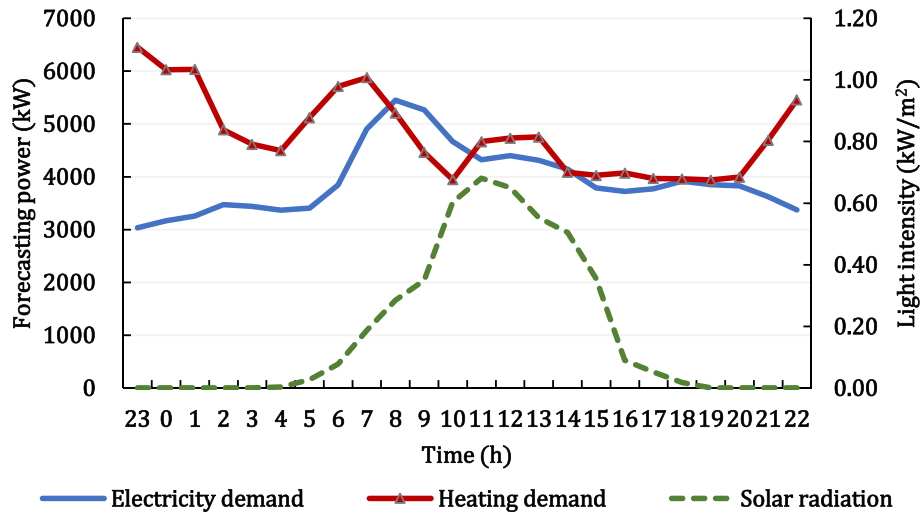


Fig. A3. Typical scenario in the heating period.

Table A1

Device parameters in the CIES.

Devices	Parameters	Value
Electric boiler system	Rated capacity of pressurized electric boiler	4100 kW
	Capacity of the hot water tank	30,000 kWh
	Maximum power of the hot water tank	4100 kW
	Heat loss rate of the hot water tank	0.001
	Efficiency of electric boiler system	0.95
Double-duty chiller	Rated capacity of the chiller	6328 kW
	Capacity of the ice storage tank	35,170 kWh
	Rated COP (cooling)	4.89
	Rated COP (ice-making)	4.03
	Heat loss rate of the ice storage tank	0.001
	Maximum power of the ice storage tank	7000 kW
GSHP	Rated capacity	2710 kW
	Rated COP (heating)	4.00
	Rated COP (cooling)	5.38
CHP	Rated capacity	5000 kW
	Gas-thermal efficiency	0.45
	Gas-electric efficiency	0.3
PV	Rated peak capacity	823 kW

Table A2

Energy price in the CIES.

Energy type	Time	Price
Electricity	Peak hours	8:00–11:00, 18:00–23:00
	Valley hours	0:00–7:00, 23:00–0:00
	Normal hours	7:00–8:00, 11:00–18:00
Natural gas	Full hours	0.29 CNY/kWh

Appendix B

The system efficiency of a CIES can be calculated as.

$$\eta_{SE} = \frac{E_{out}}{E_{in}} = \frac{E_{out}^{PL} + E_{out}^{HL} + E_{out}^{CL}}{E_{in}^{Grid} + E_{in}^{DG} + E_{in}^{GF} + E_{in}^{AH}} \quad (B1)$$

where η_{SE} is the indicator of system efficiency; E_{in} and E_{out} are the exergy input and output of the CIES within the total operation period, respectively. E_{in} consists of all the exergy inputs to the system, including the external power grid, DG, gas fuels, and ambient heat energy. E_{out} consists of all exergy consumed in the system, including the electrical, heating, and cooling loads.

The renewable energy penetration of a CIES can be calculated as.

$$\eta_{\text{REP}} = \frac{\sum R_{\text{in}}}{\sum R_{\text{end}} + \sum U_{\text{end}}} = \frac{E_{\text{in}}^{\text{DG}} + E_{\text{in}}^{\text{AH}}}{E_{\text{out}}^{\text{PL}} + E_{\text{out}}^{\text{HL}} + E_{\text{out}}^{\text{CL}}} \quad (\text{B2})$$

where η_{REP} is the indicator of renewable energy penetration; $\sum R_{\text{in}}$ is the total exergy of renewable energy that can be provided by the supply side; $\sum R_{\text{end}}$ is the total exergy of renewable energy consumed by the user side; $\sum U_{\text{end}}$ is the total exergy of nonrenewable energy consumed by the users.

The operation cost of a CIES can be calculated as.

$$C_{\text{OC}} = \sum C^k, k \in \{\text{maintenance, energy purchase, ...}\} \quad (\text{B3})$$

where C_{OC} is the total operation cost; C^k is the different types of cost considered in operation.

Appendix C

The quantitative scoring process using the modified TOPSIS method is formulated in detail as follows:

Step 1): Construct the decision matrix D .

Assume that n samples are to be evaluated in terms of three indicators: system efficiency, renewable energy penetration, and operation cost. The decision matrix $D = (r_{ij})_{(n+2) \times i}$ is constructed as follows:

$$D = \begin{bmatrix} r_{1,1} & r_{1,2} & r_{1,3} \\ \vdots & \vdots & \vdots \\ r_{n,1} & r_{n,2} & r_{n,3} \\ r_{(n+1),1} & r_{(n+1),2} & r_{(n+1),3} \\ r_{(n+2),1} & r_{(n+2),2} & r_{(n+2),3} \end{bmatrix} \quad (\text{C.1})$$

where r_{ij} is indicator j of operation sample i . When r_{ij} is taken to represent the system efficiency and renewable energy penetration, it can be calculated directly by applying Equations (B.1) and (B.2). When r_{ij} represents the operational cost, it is obtained by performing a monotonically increasing process to the calculation result of Equation (B.3). $r_{(n+1),j}$ and $r_{(n+2),j}$ are defined as positive and negative ideal solutions, respectively. $r_{(n+1),j}$ can be obtained by solving the optimization problems in Section 3.1, whereas $r_{(n+2),j}$ are all set to 0.

Step 2): Normalize the decision matrix D .

Equation (C.2) is used to normalize all elements in matrix D to obtain the dimensionless normalized decision matrix D' .

$$r'_{ij} = \frac{r_{ij} - r_{(n+2),j}}{r_{(n+1),j} - r_{(n+2),j}} \quad (\text{C2})$$

$$D' = \begin{bmatrix} r'_{1,1} & r'_{1,2} & r'_{1,3} \\ \vdots & \vdots & \vdots \\ r'_{n,1} & r'_{n,2} & r'_{n,3} \\ r'_{(n+1),1} & r'_{(n+1),2} & r'_{(n+1),3} \\ r'_{(n+2),1} & r'_{(n+2),2} & r'_{(n+2),3} \end{bmatrix} \quad (\text{C3})$$

where r'_{ij} is the normalised indicator j of operation sample i .

Step 3): Assign the indicator weights.

The weight of an indicator refers to its relative importance in the overall evaluation. In this study, the weights of the three indicators are set as ω_j , and $\sum_{j=1}^3 \omega_j = 1$.

Step 4): Obtain the distances of existing samples from the ideal solutions.

Euclidean distance is used to calculate the distance between the operational and ideal indicators of the sample. The distances of each sample to the positive ideal solution and negative ideal solution are denoted by d_i^+ and d_i^- , respectively, which can be calculated as follows:

$$d_i^+ = \sqrt{\sum_{j=1}^3 \omega_j (r'_{ij} - r'_{(n+1),j})^2}, i = 1, 2, \dots, n \quad (\text{C4})$$

$$d_i^- = \sqrt{\sum_{j=1}^3 \omega_j (r'_{ij} - r'_{(n+2),j})^2}, i = 1, 2, \dots, n \quad (\text{C5})$$

In addition, Equations (C.4) and (C.5) can be used to evaluate the performance of a single indicator by neglecting the summation and weight coefficients.

Step 5): Calculate the quantitative scores for operation performance.

Depending on the distances obtained in Step 4), the relative closeness R_i of each sample to the ideal solution is calculated according to Equation (C.6). Then, the evaluation score S_i is obtained using Equation (C.7), where a larger value indicates a better operational performance.

$$R_i = \frac{d_i^-}{d_i^+ + d_i^-}, i = 1, 2, \dots, n \quad (C6)$$

$$S_i = 100 \times R_i, i = 1, 2, \dots, n \quad (C7)$$

Appendix D

The daily electrical heating/cooling ratio λ can be calculated using Equations (D.1) or (D.2), which represent the features of the daily load level.

$$\lambda_i^{\text{EH}} = \frac{\sum_{t=1}^T Q_{i,t}^{\text{E}}}{\sum_{t=1}^T Q_{i,t}^{\text{H}}} \quad (D1)$$

$$\lambda_i^{\text{EC}} = \frac{\sum_{t=1}^T Q_{i,t}^{\text{E}}}{\sum_{t=1}^T Q_{i,t}^{\text{C}}} \quad (D2)$$

where $Q_{i,t}^{\text{E}}$, $Q_{i,t}^{\text{H}}$ and $Q_{i,t}^{\text{C}}$ represent the electricity, heating, and cooling loads of sample i at time t , respectively; λ_i^{EH} and λ_i^{EC} represent the daily electrical heating ratio and the daily electrical cooling ratio of sample i , respectively; T is the time period.

The daily load baseline ratio is a reference base related to the maximum heating/cooling load for a single operating day. In this study, the maximum heating/cooling load for a single day is used as the baseline.

The load distribution ξ can be calculated using Equation (D.3), which represents the distribution properties of the normalized load profiles.

$$\xi_{i,t}^{\Omega} = \frac{Q_{i,t}^{\Omega}}{\sum_{t=1}^T Q_{i,t}^{\Omega}}, \Omega \in \{\text{electricity, heating, cooling}\} \quad (D3)$$

where $Q_{i,t}^{\Omega}$ represents the load of sample i at time t and $\xi_{i,t}^{\Omega}$ represents the load distribution of sample i at time t .

Appendix E

The mean absolute percentage error M_{APE} and mean accuracy M_{A} are calculated as follows:

$$M_{\text{APE}} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - y_i'}{y_i} \right| \times 100\% \quad (E1)$$

$$M_{\text{A}} = 1 - M_{\text{APE}} \quad (E2)$$

$$W_{\text{MA}} = \alpha_1 M_{\text{A},1} + \dots + \alpha_j M_{\text{A},j} \quad (E3)$$

where y_i' and y_i are the indicator values solved by the surrogate model and optimization for the i th sample, respectively; $M_{\text{A},j}$ is the mean accuracy of j th optimal indicator; and α_j is the weight of j th optimal indicator, and the indicators are set with the same weights in this study.

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